

ON3VZ

SWR sweep per band with the NanoVNA-H4 and NanoVNA-Saver

Step-by-step plan: install the software, calibrate, and measure every band from 160 m to 6 m

Version 1.2 · English edition · October 2026 · companion to *Measuring as a radio amateur with the NanoVNA-H4 and the tinySA*

on3vz.github.io

How to use this plan. Work through parts A and B once. After that, Part C is the routine you repeat for every band, and Part D gives you the quick all-bands check.

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0. Before you start

This document is a practical, executable plan: install the software, calibrate, and then measure band by band. For the background (what SWR is, how to read a Smith chart, how the NanoVNA itself works) it refers to the main manual, *Measuring as a radio amateur with the NanoVNA-H4 and the tinySA*. Here you only find what you actually have to do, in order.

SAFETY

Discharge the antenna before you connect it, and disconnect the transceiver from the antenna line. See the main manual, sections 0.3 and 0.4, for exactly how.

0.1 What you need

- A NanoVNA-H4, charged, with the supplied SMA cable, the SMA female-female coupler and the open/short/load caps (all in the box).
- An SMA to SO-239 adapter (for your HF antennas with a PL connector). Its SMA side can be male or female; B.1 explains what that changes.
- A Windows PC with a free USB port.
- The USB-C cable from the NanoVNA's box.

0.2 Overview of this plan

Part	Contents
A	Installing NanoVNA-Saver and connecting the NanoVNA
B	Calibrating in the software: per band, or one wide calibration
C	SWR sweep per band, 160 m to 6 m, each with a band sheet, plus a checklist for every result
D	All bands in one sweep
E	Saving results for your log

Part A. Installing the software and connecting

A.1 Downloading NanoVNA-Saver

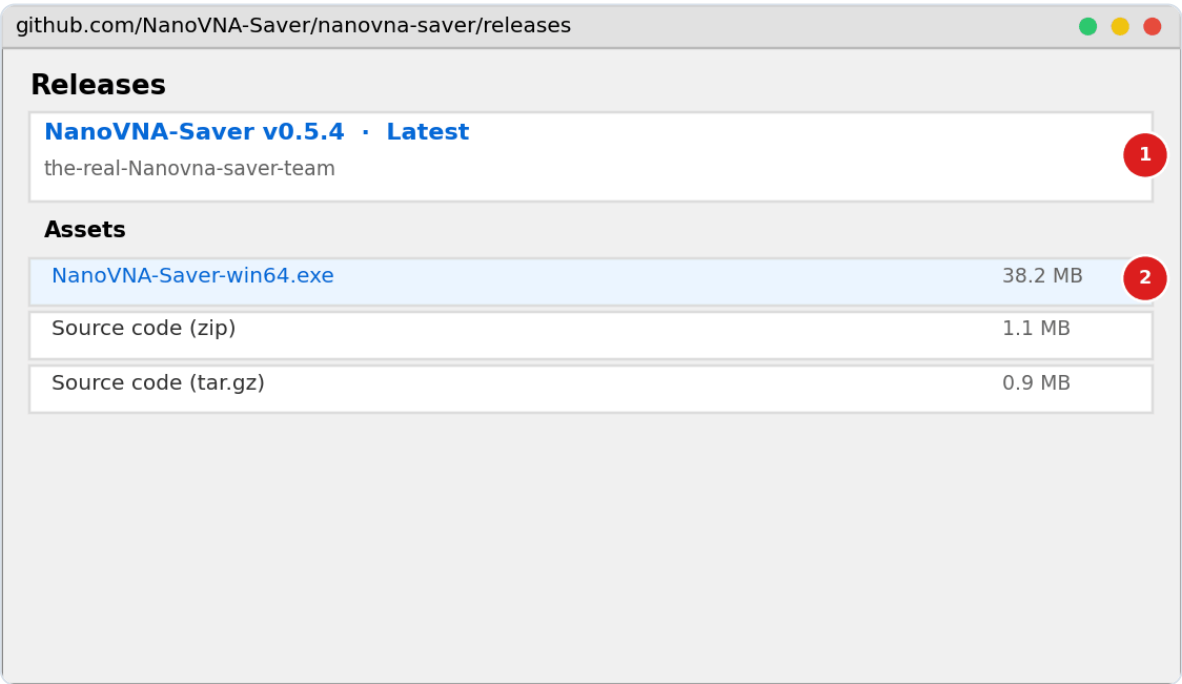


Figure 1 · The download page on GitHub (illustration; your version number will be newer).

No.	What you see here
1	The newest version (marked Latest) is at the top. Always use this one unless you have a reason to take an older one.
2	Under Assets: the file ending in <code>.exe</code> is the Windows version you need.

Step 1 Go to github.com/NanoVNA-Saver/nanovna-saver/releases in your browser.

Step 2 Under **Assets** of the top (newest) release, click the Windows file ending in `.exe`.

Step 3 Your browser downloads the file to your Downloads folder. Wait until the download has completely finished (the progress bar in your browser disappears).

A.2 Starting it and the security warning

NanoVNA-Saver is free, open-source software without a digital signature from a large publisher. That is why Windows shows a warning the first time. It is normal for this kind of program and not an error.

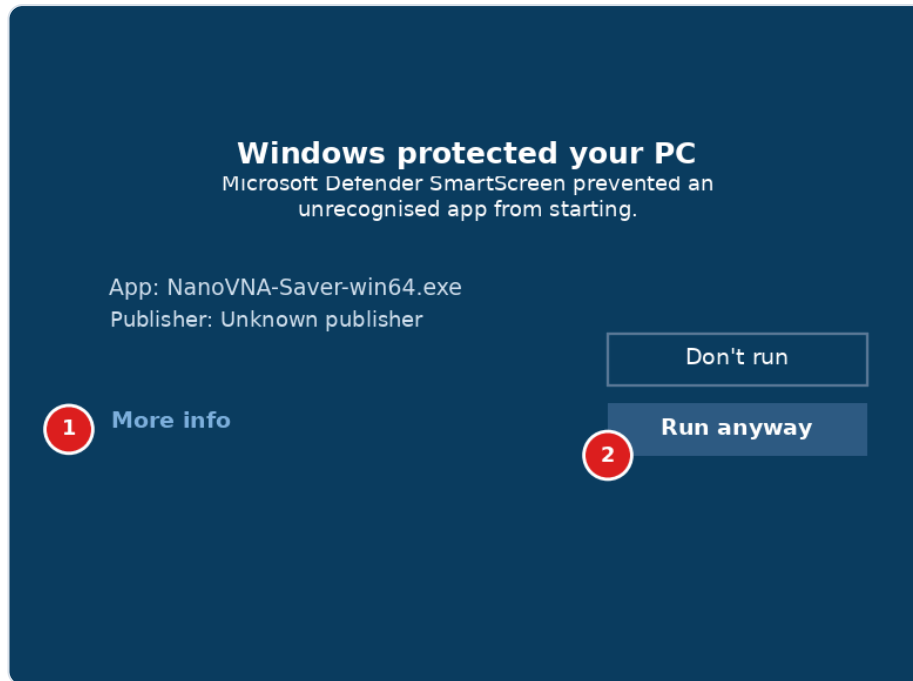


Figure 2 · The Windows security warning (SmartScreen).

No.	What you see here
1	Click More info . The warning expands and shows extra buttons.
2	Click Run anyway . This starts the program.

Step 1 Double-click the downloaded file in your Downloads folder.

Step 2 If the blue "Windows protected your PC" warning appears, click **More info**.

Step 3 Click **Run anyway**.

Step 4 The program opens straight away. There is no installation wizard: NanoVNA-Saver does not need to be installed, it runs directly from the downloaded file.

Step 5 Tip: move the `.exe` file to a fixed folder (e.g. `Documents`) so you can find it easily. Put a shortcut on your desktop if you like.

CAUTION

If you get no warning at all, or your antivirus blocks the file completely: that can happen on some company computers. Ask your IT department for an exception, or use a private computer.

A.3 Connecting and recognising the NanoVNA

Step 1 Plug the USB-C cable into the NanoVNA and into a free USB port on your PC.

Step 2 Switch the NanoVNA on with the slide switch.

Step 3 Open Device Manager: right-click the Windows Start button and choose **Device Manager**.

Step 4 Click the arrow in front of **Ports (COM & LPT)**. A line **STMicroelectronics Virtual COM Port** appears with a number, for example COM4. Remember this number.

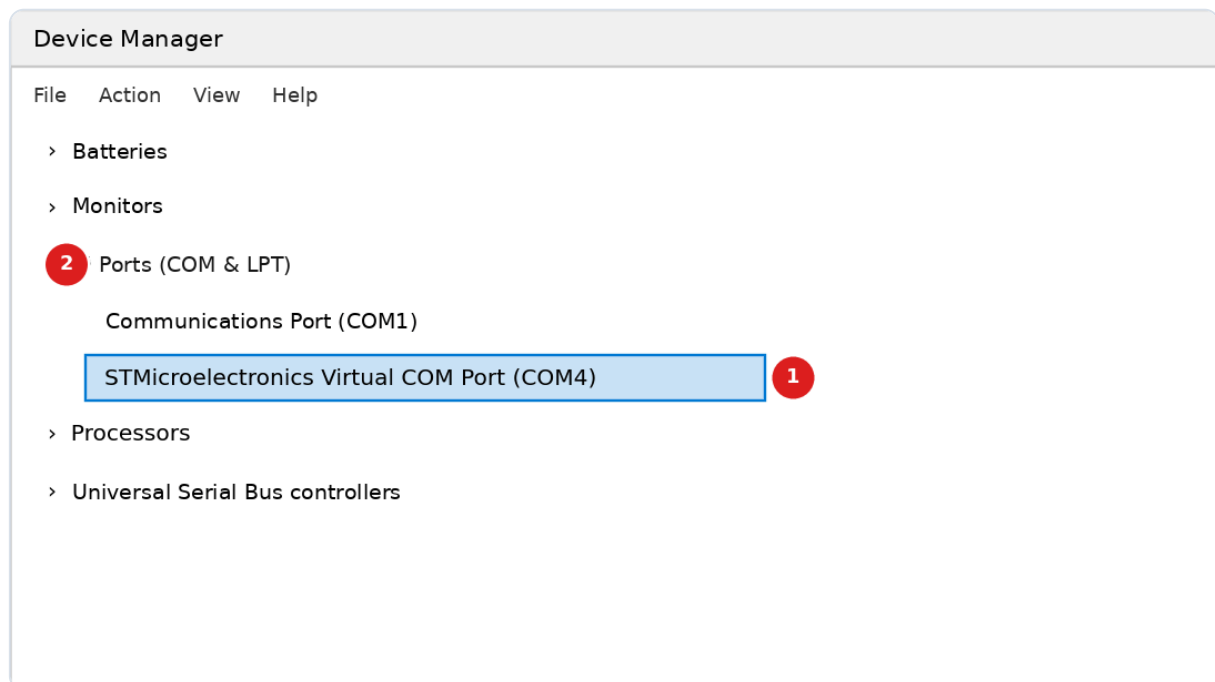


Figure 3 · Device Manager after connecting.

No. What you see here

1 The NanoVNA, recognisable as STMicroelectronics Virtual COM Port. You will need the number (here COM4) in a moment.

2 Look under this heading, not under any other.

What you see	What it means	What you do
No new line in Device Manager	The cable carries no data, or the driver is missing	Use the cable from the NanoVNA's box, try another (preferably USB 2.0) port
Yellow exclamation mark on the device	Driver not installed correctly	Right-click the line > Update driver

A.4 First test connection in NanoVNA-Saver

Step 1 Open NanoVNA-Saver (double-click the .exe file or the shortcut).

Step 2 Bottom left, under **Serial port control**: click the drop-down and choose the COM number you noted in A.3.

Step 3 Click **Connect to device**. The button changes to **Disconnect** as soon as the connection works.

Step 4 Under **Sweep control**, top left, enter Start **6.9M** and Stop **7.3M**, and click **Sweep**.

Step 5 If graphs appear on the right of the window, the connection works. Continue with Part B.

What you see	What it means	What you do
Connect to device stays grey or gives an error	Wrong COM port chosen, or the unit is not on	Check that the NanoVNA is on and choose the right port again
No graph after Sweep	The software is not really talking to the unit. Even with nothing connected to the port you would still get a line, so no line at all means no data	Check that the button reads Disconnect (connected), then click Sweep again

Part B. Calibrating in the software

Calibration teaches the software what a "perfect open", a "perfect short" and a "perfect 50 ohm load" look like at the point where you screw on the caps, for the frequency range you set. This part first shows what screws onto what, then two approaches: one calibration per band (most accurate) or one wide calibration for all HF bands together (quickest).

B.1 The caps, the coupler and where to calibrate

Golden rule: calibrate exactly where the antenna will be connected afterwards. Everything between the NanoVNA and that point (test cable, coupler) must be in place while you calibrate and stay in place while you measure. Calibrating with the caps directly on the NanoVNA port and then measuring through the test cable is wrong: the cable is not in the calibration. Caps on the port and the antenna adapter on the port as well is fine: then both use the same point.

The NanoVNA port is female (a hole in the middle), the test cable is male at both ends (a pin), and the caps are male too. Male does not fit male, so the caps go on the **SMA female-female coupler** at the end of the test cable. The coupler stays on during the whole calibration; only the cap is swapped.

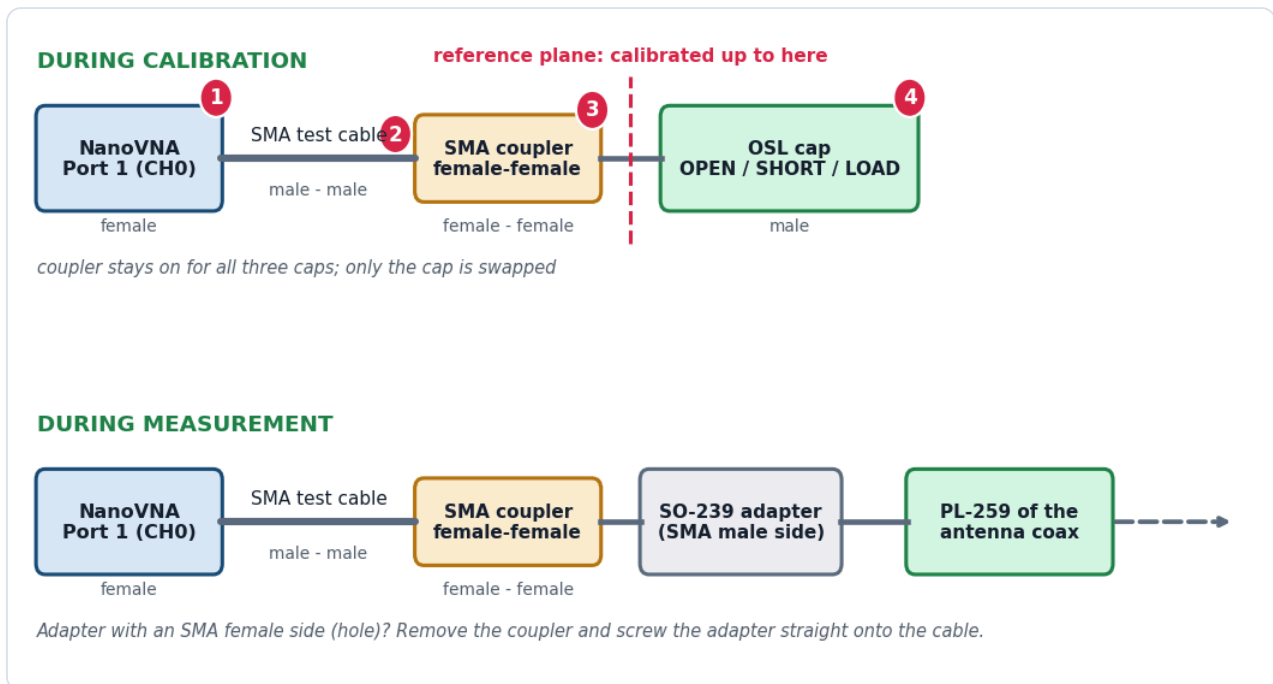


Figure 4 · Top: during calibration. Bottom: while measuring.

After calibrating: an SO-239 adapter with an SMA male side (pin) goes onto the coupler. One with an SMA female side (hole) goes straight onto the cable; remove the coupler first (on HF that centimetre makes no visible difference).



Figure 5 · Recognising the caps: look into them from the front.

No.	Cap	What you see inside	How to remember it
1	OPEN	Only white plastic, no metal, no pin	"Open": nothing connected, only white
2	SHORT	All metal, no white, no pin	"Short": shorted, one piece of metal
3	LOAD	White plastic with a small thin pin in the middle	"Load": a real load, with a mini pin

No marking? Shine a torch into the cap. Only white: OPEN. A small pin: LOAD. Only metal: SHORT.

B.2 Choosing an approach

Approach	Advantage	Disadvantage	Recommended for
One calibration per band	Most accurate within that band	You have to calibrate again for each band	Fine tuning, reading the Smith chart
One wide calibration, 1 to 30 MHz	Quick: do it once for all HF bands	Slightly less accurate at the edges of the range	Quickly checking all bands (Part D)

B.3 Calibrating with the Calibration assistant

This example shows the calibration for 40 m (6.9 to 7.3 MHz). For any other band you repeat the same steps with the range from the band sheet in Part C.

Step 1 Enter Start and Stop for the band you want to calibrate (for 40 m: **6.9M** and **7.3M**). Always end with the letter **M**.

Step 2 **Segments:** leave at 1 for every band up to 0.6 MHz wide; set 2 for 6 m (and optionally 10 m). Each segment is 101 measuring points.

Step 3 Click **Calibration**, bottom left. Click **Reset** (if present) to clear an old calibration.

Step 4 Click **Calibration assistant**.

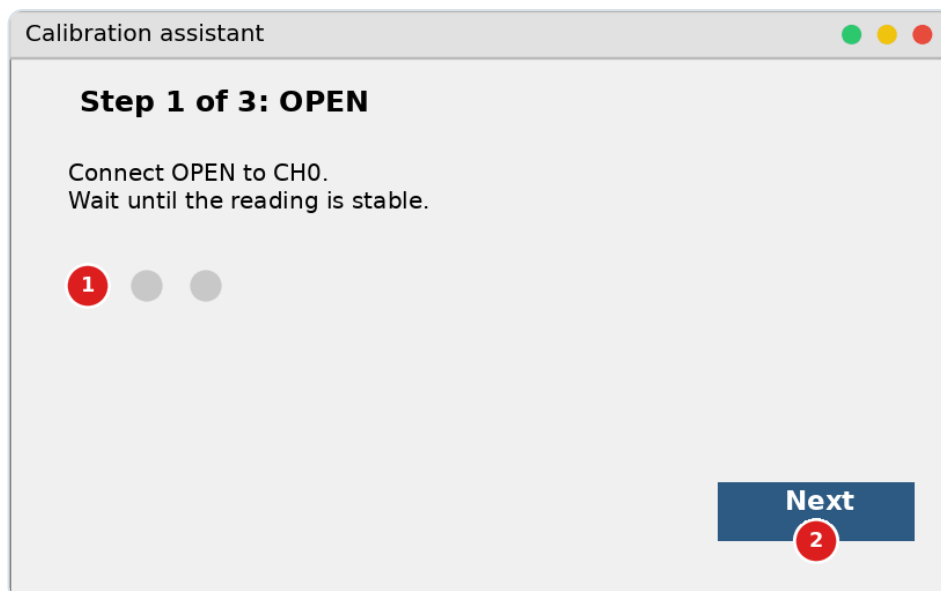


Figure 6 · Step 1 of the assistant, OPEN (simplified illustration).

No.	What you see here
1	The step you are on now.
2	Click Next only once the reading on screen is stable (about 2 seconds).
Step 1	Screw the cap the assistant asks for (here OPEN) onto the coupler at the end of your SMA cable (B.1). Wait 2 seconds until the reading is stable and click Next.
Step 2	Swap to the next cap it asks for (SHORT). Wait and click Next. The coupler stays on.
Step 3	Swap to LOAD . Wait and click Next.
Step 4	Asked for Through ? For an antenna (Port 1 only) click Cancel . Only for filters and chokes, with both cable ends joined by the coupler, do you complete it.

The assistant may ask in the order short, open, load or open, short, load. Both are fine, as long as the cap on the coupler is the one it asks for.

USE THE ASSISTANT, NOT THE LOOSE BUTTONS

The Calibration window also has separate **Open**, **Short**, **Load** and **Through** buttons. Do not use them: the program itself warns that they do not run a sweep and do not interact with the NanoVNA's own calibration.

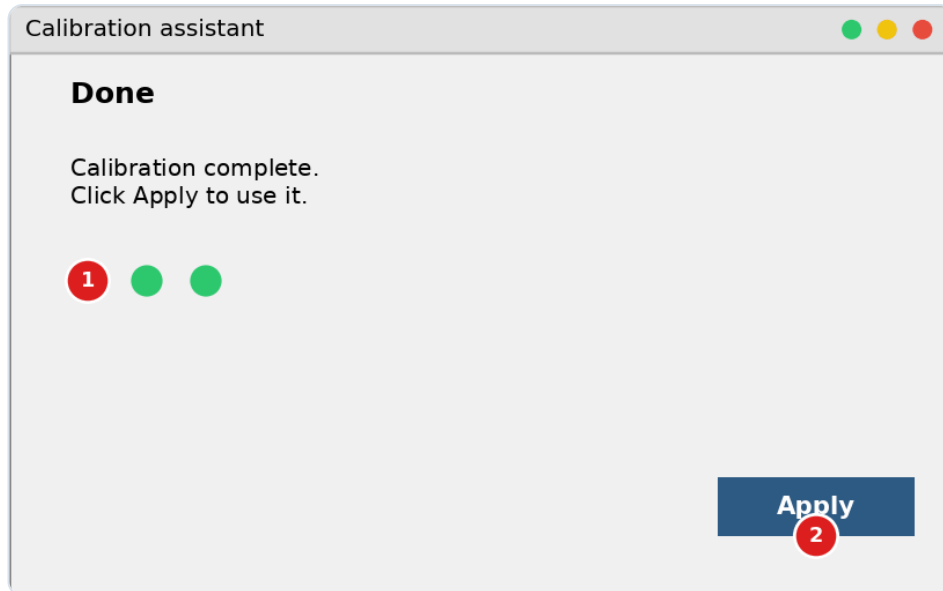


Figure 7 · The assistant has finished.

Step 1 Click **Apply**. The calibration is now active for this frequency range.

B.4 Saving the calibration as a file

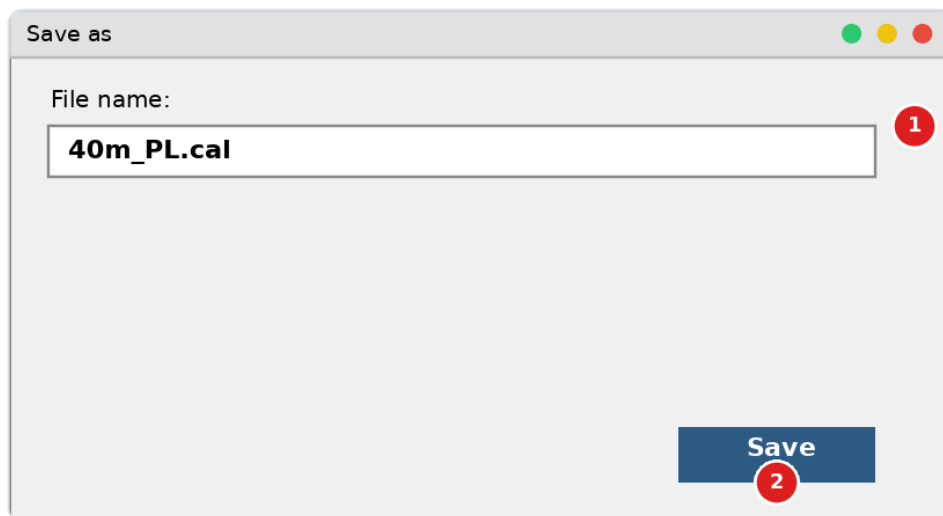


Figure 8 · Use a clear file name.

No.	What you see here
1	File name with band and connection , for example <code>40m_PL.cal</code> . You will recognise the file straight away later.
2	Save confirms and closes the window.

Step 1 Under Calibration, click **Save**.

Step 2 Type a clear name following the pattern `<band>_<connection>.cal`, for example `40m_PL.cal`.

Step 3 Choose a fixed folder, for example `Documents`, and click Save.

Step 4 Repeat B.3 and B.4 for every band in Part C, each time with that band's start and stop frequency.

TIP

Create the folder **Documents** before you start. That way all your **.cal** files end up in one place and are easy to find.

B.5 One wide calibration for all HF bands

Step 1 Enter Start **1M** and Stop **30M**.

Step 2 Increase **Segments** to at least 8 (that gives 808 measuring points instead of 101, sharp enough for all bands together).

Step 3 Run the Calibration assistant as in B.3 with this wide range.

Step 4 Save it as **HF_1-30MHz_PL.cal** (B.4).

B.6 Loading a saved calibration

Step 1 Under Calibration, click **Load**.

Step 2 Browse to **Documents** and double-click the **.cal** file you want.

Step 3 Check that Start and Stop at the top automatically show the values of that calibration.

CAUTION

A calibration is only valid for the frequency range in which it was made. Load a 40 m calibration and measure on 20 m, and the measurement cannot be trusted. Every band sheet in Part C tells you which file to load.

Part C. SWR sweep per band

For every band you follow the same fixed method. First the method itself, then a sheet per band with the right settings.

C.1 The fixed method (for every band)

- Step 1** Disconnect the transceiver from the antenna switch.
- Step 2** Discharge the coax: touch the centre pin of the PL-259 to the metal nut for 2 seconds.
- Step 3** Set the antenna switch to the antenna you want to measure.
- Step 4** In NanoVNA-Saver, enter Start and Stop from the band's sheet (or Center and Span: one pair, never both).
- Step 5** Under Calibration, click **Load** and choose that band's `.cal` file (or the wide HF file).
- Step 6** Check that Start and Stop still show the band's values after loading.
- Step 7** Connect the antenna coax exactly as when you calibrated (B.1): SMA cable, coupler if your adapter needs it, SO-239 adapter.
- Step 8** Click **Sweep**. The software only measures when you click: after every change, click Sweep again.
- Step 9** Run the checklist in C.16.
- Step 10** Click the lowest point of the curve in the VSWR graph to put a marker on it, or read the value in the Markers table on the left.
- Step 11** Write down the frequency and the SWR value at the lowest point.
- Step 12** Compare with the "Expected dip" row in the band sheet and with the table "What if I see this" in C.15.
- Step 13** Save the result (see Part E) before you move on to the next band.

On the PC there is no one-click RECALL as on the unit: Start/Stop and loading the `.cal` file are always two separate actions. Which of these bands you may transmit on depends on your licence class.

C.2 Band 160 m

Setting	Value
START / STOP	1.8M / 2.0M
or CENTER / SPAN	1.9M / 0.2M
Segments	1
Calibration file	160m_PL.cal
Expected dip	inside 1.81 to 2.0 MHz

C.3 Band 80 m

Setting	Value
START / STOP	3.5M / 3.8M
or CENTER / SPAN	3.65M / 0.3M
Segments	1
Calibration file	80m_PL.cal
Expected dip	around 3.65 MHz (depends on the antenna length)

C.4 Band 60 m

Setting	Value
START / STOP	5.3M / 5.4M
or CENTER / SPAN	5.35M / 0.1M
Segments	1
Calibration file	60m_PL.cal
Expected dip	around 5.36 MHz (check your national allocation)

C.5 Band 40 m

Setting	Value
START / STOP	6.9M / 7.3M
or CENTER / SPAN	7.1M / 0.4M
Segments	1
Calibration file	40m_PL.cal
Expected dip	around 7.1 MHz

C.6 Band 30 m

Setting	Value
START / STOP	10.0M / 10.2M
or CENTER / SPAN	10.1M / 0.2M
Segments	1
Calibration file	30m_PL.cal
Expected dip	around 10.12 MHz

CAUTION

30 m is a narrow band: only 50 kHz (10.100 to 10.150 MHz); the sheet sweeps 200 kHz around it. With 101 points that is a point every 2 kHz, sharp enough with one segment. Calibrate extra carefully here.

C.7 Band 20 m

Setting	Value
START / STOP	13.9M / 14.5M
or CENTER / SPAN	14.2M / 0.6M
Segments	1
Calibration file	20m_PL.cal
Expected dip	around 14.15 MHz

C.8 Band 17 m

Setting	Value
START / STOP	18.0M / 18.2M
or CENTER / SPAN	18.1M / 0.2M
Segments	1
Calibration file	17m_PL.cal
Expected dip	around 18.12 MHz

C.9 Band 15 m

Setting	Value
START / STOP	20.9M / 21.5M
or CENTER / SPAN	21.2M / 0.6M
Segments	1
Calibration file	15m_PL.cal
Expected dip	around 21.2 MHz

C.10 Band 12 m

Setting	Value
START / STOP	24.8M / 25.0M
or CENTER / SPAN	24.9M / 0.2M
Segments	1

Setting	Value
Calibration file	12m_PL.cal
Expected dip	around 24.94 MHz

C.11 Band 11 m (CB, check only)

Setting	Value
START / STOP	26.9M / 27.4M
or CENTER / SPAN	27.15M / 0.5M
Segments	1
Calibration file	11m_PL.cal
Expected dip	CB band, not an amateur band: only to check a CB antenna

C.12 Band 10 m

Setting	Value
START / STOP	28.0M / 29.7M
or CENTER / SPAN	28.85M / 1.7M
Segments	1 or 2
Calibration file	10m_PL.cal
Expected dip	depends strongly on the antenna; multiband antennas are often tuned broader here

TIP

10 m is 1.7 MHz wide. Use 2 segments, or if your antenna is narrowband here, measure in two steps, 28.0M to 28.7M and 28.7M to 29.7M, for a sharper picture of each part.

C.13 Band 6 m

Setting	Value
START / STOP	50.0M / 52.0M
or CENTER / SPAN	51.0M / 2.0M
Segments	2
Calibration file	6m_PL.cal
Expected dip	around 50.5 MHz

C.14 Overview table

Band	START	STOP	CENTER	SPAN	Segments	Calibration file
160 m	1.8M	2.0M	1.9M	0.2M	1	160m_PL.cal
80 m	3.5M	3.8M	3.65M	0.3M	1	80m_PL.cal
60 m	5.3M	5.4M	5.35M	0.1M	1	60m_PL.cal
40 m	6.9M	7.3M	7.1M	0.4M	1	40m_PL.cal
30 m	10.0M	10.2M	10.1M	0.2M	1	30m_PL.cal
20 m	13.9M	14.5M	14.2M	0.6M	1	20m_PL.cal
17 m	18.0M	18.2M	18.1M	0.2M	1	17m_PL.cal
15 m	20.9M	21.5M	21.2M	0.6M	1	15m_PL.cal
12 m	24.8M	25.0M	24.9M	0.2M	1	12m_PL.cal
11 m	26.9M	27.4M	27.15M	0.5M	1	11m_PL.cal
10 m	28.0M	29.7M	28.85M	1.7M	1 or 2	10m_PL.cal
6 m	50.0M	52.0M	51.0M	2.0M	2	6m_PL.cal

CENTER and SPAN are just another way to set the same range: CENTER is the middle frequency, SPAN the total width. Fill in Start/Stop and the software calculates Center/Span, and the other way round. Use one pair per band, never both one after the other.

C.15 What if I see this (applies to every band)

What you see	What it means	What you do
Dip above the band	Antenna electrically too short for this band	Lengthen the wire, measure again
Dip below the band	Antenna electrically too long	Shorten the wire in small steps, measure again
Dip in the band, SWR 1.0 to 1.5	Good	Nothing; save the result (Part E)
Dip in the band, SWR above 2	Resonant, but the wrong impedance	Check the balun or matching (see the main manual, Smith chart)
No dip, high and flat everywhere	Open circuit, a cap still on instead of the antenna, or the wrong calibration loaded	Check which <code>.cal</code> file is loaded, check the coax
A jagged, jumpy line	Wrong graph type, a loose connector, or interference	Check the graph, tighten the connectors, measure again
The curve looks different from the last measurement of the same band	Calibration for a different range loaded, or the antenna has physically changed	Load the right <code>.cal</code> file and measure again

C.16 Is my result right? The checklist after every sweep

No.	Check	Good	Not good: what to do
1	Graph type. Are you reading the VSWR graph?	VSWR	Another graph (phase, log mag): read the VSWR graph
2	Range. Do Start and Stop show the band you think?	The right band	Enter the band's values and load the matching <code>.cal</code>
3	Calibration. Is the right <code>.cal</code> file loaded for this range?	File name and range match	Load the right file (B.6)
4	Shape of the curve	A smooth V-shaped dip	Flat and high: open connection, cap still on, or wrong calibration. Jagged: wrong graph, loose connector or interference
5	LOAD test. Screw the LOAD cap where the antenna goes and click Sweep	A flat line at SWR 1.0 (for example 1.002)	Clearly higher: the calibration itself is wrong. Calibrate again

Part D. All bands in one sweep

Handy for seeing at a glance whether all bands of a multiband antenna (a fan dipole, a trap dipole, an EFHW or a multiband vertical) are still where they should be, without switching band by band.

NOT FOR LINKED DIPOLES

A linked dipole such as the SOTabeams Band Hopper only resonates on the band its links are set for. You will not see all its bands in one sweep: set the links for a band, sweep that band with Part C, and repeat.

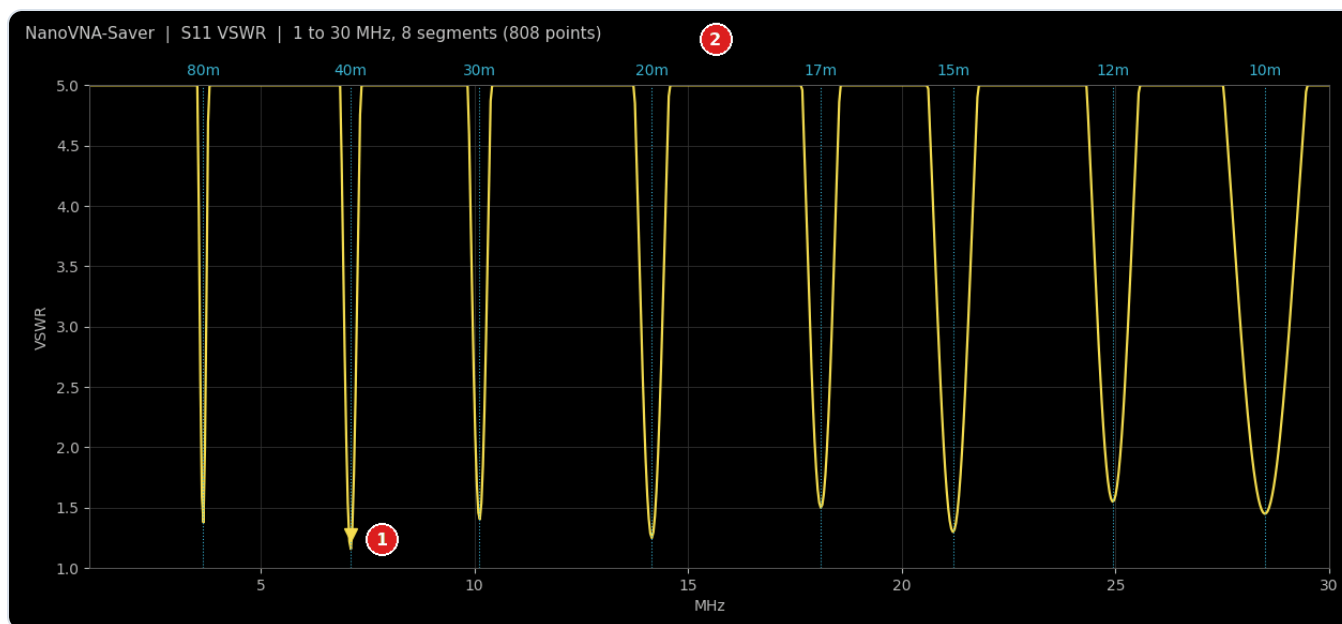


Figure 9 · 1 to 30 MHz in one sweep, with band markers.

No.	What you see here
1	The 40 m dip, with a marker placed on it.
2	Title bar: confirms the range (1 to 30 MHz) and the number of segments (8, good for 808 measuring points).

Step 1 Disconnect the transceiver and discharge the coax (C.1).

Step 2 Enter Start 1M and Stop 30M.

Step 3 Increase Segments to 8.

Step 4 Load the wide calibration: Calibration > Load > HF_1-30MHz_PL.cal (see B.5).

Step 5 Connect the antenna and click Sweep.

Step 6 Click each dip you expect (80, 40, 30, 20, 15, 10 m) to put a marker on it.

Step 7 Check per band: does the dip lie within the band limits from the sheets in Part C?

Step 8 A band without a clear dip? Zoom in by setting Start and Stop to just that band, load the matching .cal from Part C, and measure again as in Part C.

TIP

Click **Set current as reference** to keep this wide measurement as a comparison line. Measure again later and you see immediately whether anything in your antenna setup has changed.

Part E. Saving results

E.1 File naming convention

Use the same pattern for every saved measurement, so your files automatically end up in a logical order:

`YYYY-MM-DD_antenna_band.s1p`

For example: `2026-09-26_EFW_40m.s1p`

E.2 Saving a measurement as S1P

Step 1 After a successful sweep: click **Files** on the left, then **Save 1-Port file (S1P)**.

Step 2 Browse to (or create) the folder **Documents**.

Step 3 Type the file name following E.1 and click Save.

Step 4 Repeat this for every band you have measured.

E.3 A screenshot for your log

Step 1 Press **Windows** + **Shift** + **S**.

Step 2 Drag a frame around the NanoVNA-Saver window.

Step 3 Open Paint (or another program), paste the image (**Ctrl** + **V**) and save it under the same name as the S1P file, but as **.png**.

E.4 Loading a measurement again later and comparing

Step 1 Click **Files**, then **Load as sweep**, and choose a previously saved **.s1p** file.

Step 2 The measurement appears in the graphs as if you had just measured it. Use **Load reference** instead to show it as a fixed comparison line next to a new sweep.

TIP

Keep this document and your **.cal** files together in **Documents**. A software calibration corrects whatever the instrument sends to the PC, so keep the instrument's own calibration state the same as when you made the **.cal** file. The simplest habit: make software calibrations with the instrument's calibration cleared (**CALIBRATE > RESET**) and always measure that way. Files made like this remain valid after a firmware update of the NanoVNA (main manual, 8.8).